STATS 210 Assignment 1 Solutions Semester 2, 2017

1a) $\mathbb{P}(H) = 0.5$
$\mathbb{P}(D) = 0.3$
$\mathbb{P}(S) = 0.2$

$\mathbb{P}(C \mid H) = 0.6$
$\mathbb{P}(C \mid D) = 0.8$
$\mathbb{P}(C \mid S) = 0$

b) i) H, D, and S for Ω : yes.

ii) C and $\bar{C}$ for H : no. $(C \cup \bar{C} = \Omega \neq H)$.

iii) $C \cap H$ and $\bar{C} \cap H$ for H : yes.

iv) $C \cap H$ and $\bar{C} \cap H$ for C : no. (Their union is H, not C.)

c) $\mathbb{P}(C) = \mathbb{P}(C|H)\mathbb{P}(H) + \mathbb{P}(C|D)\mathbb{P}(D) + \mathbb{P}(C|S)\mathbb{P}(S)$
because H, D, and S partition Ω

$= 0.6 * 0.5 + 0.8 * 0.3 + 0 * 0.2$

$\underline{\mathbb{P}(C) = 0.54}$

d) $\mathbb{P}(D \mid C) = \dfrac{\mathbb{P}(C|D)\mathbb{P}(D)}{\mathbb{P}(C)}$ (Bayes Thm)

$= \dfrac{0.8 * 0.3}{0.54}$ using (c)

$\underline{\mathbb{P}(D \mid C) = \dfrac{4}{9} \quad (0.44)}$.

-probability Sally was taken dancing given that she calls him back.

1e) $\mathbb{P}(D \cap K) = 0.24$

$\mathbb{P}(K \mid H) = 0.10$

$\mathbb{P}(K \mid S) = 0$

$\mathbb{P}(C \mid K) = 0.7$

f) $\mathbb{P}(K \mid D) = \dfrac{\mathbb{P}(K \cap D)}{\mathbb{P}(D)}$

$= \dfrac{0.24}{0.3}$ using (e) and (a)

$\underline{\mathbb{P}(K \mid D) = 0.8}$

-probability that a dancing date keels over.

g) $\mathbb{P}(K) = \mathbb{P}(K|D)\mathbb{P}(D) + \mathbb{P}(K|H)\mathbb{P}(H) + \mathbb{P}(K|S)\mathbb{P}(S)$
because D, H, and S partition Ω

$= 0.8 * 0.3 + 0.10 * 0.5 + 0 * 0.2$

$\underline{\mathbb{P}(K) = 0.29}$

h) {Disaster} $= K \cup \bar{C}$.

$\therefore \mathbb{P}(\text{Disaster}) = \mathbb{P}(K \cup \bar{C})$

$= \mathbb{P}(K) + \mathbb{P}(\bar{C}) - \mathbb{P}(\bar{C} \cap K)$. (a)

Now $\mathbb{P}(K) = 0.29$ from (g)
and $\mathbb{P}(\bar{C}) = 1 - \mathbb{P}(C) = 1 - 0.54 = 0.46$ using (c).

Finally $\mathbb{P}(\bar{C} \cap K) = \mathbb{P}(\bar{C} \mid K)\mathbb{P}(K)$
$= \{1 - \mathbb{P}(C|K)\}\mathbb{P}(K)$
$= \{1 - 0.7\} * 0.29$ using (e) and (g)
$= 0.087$.

1h cont.) So $\mathbb{P}(\text{disaster}) = 0.29 + 0.46 - 0.087$

$$\mathbb{P}(\text{disaster}) = 0.663$$

bad news for Tom!

2a) $\mathbb{P}(D) = 0.5$, ①

$\mathbb{P}(S) = 0.5$, ②

and $\mathbb{P}(A|D) = 2\,\mathbb{P}(A|S)$. ③

b) Amy needs to calculate $\mathbb{P}(D|A)$ (or alternatively $\mathbb{P}(S|A)$)

Now $\mathbb{P}(D|A) = \dfrac{\mathbb{P}(A|D)\,\mathbb{P}(D)}{\mathbb{P}(A)}$ Bayes Rule

$$= \frac{\mathbb{P}(A|D)\,\mathbb{P}(D)}{\mathbb{P}(A|D)\mathbb{P}(D) + \mathbb{P}(A|S)\mathbb{P}(S)}$$

$$= \frac{\mathbb{P}(A|D) * 0.5}{\mathbb{P}(A|D) * 0.5 + \mathbb{P}(A|S) * 0.5}$$ using ① and ②

$$= \frac{\mathbb{P}(A|D)}{\mathbb{P}(A|D) + \mathbb{P}(A|S)}$$ cancelling

$$= \frac{2\,\mathbb{P}(A|S)}{2\,\mathbb{P}(A|S) + \mathbb{P}(A|S)}$$ using ③

$$= \frac{2}{2+1}$$ cancelling $\mathbb{P}(A|S)$

$$\therefore \mathbb{P}(D|A) = \frac{2}{3}$$

- the probability of having a dissimilar immune system, given that the scent is attractive.

This means $\mathbb{P}(S|A) = \frac{1}{3}$.

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3)a) Class 1 : $x=1$ so $Y_1 \sim \text{Binomial}(n=10, p=0.3)$

Class 2 : $x=2$ so $Y_2 \sim \text{Binomial}(n=10, p=0.6)$

Class 3 : $x=3$ so $Y_3 \sim \text{Binomial}(n=10, p=0.9)$.

3b) $Y_2 \sim \text{Binomial}(n=10, p=0.6)$, so :

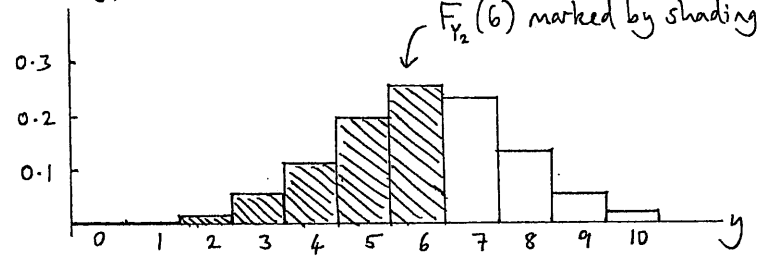
$$p_4 = \mathbb{P}(Y_2 = 4) = \binom{10}{4}(0.6)^4(0.4)^6$$

$$\Rightarrow p_4 = 0.11$$

$$p_5 = \mathbb{P}(Y_2 = 5) = \binom{10}{5}(0.6)^5(0.4)^5$$

$$\Rightarrow p_5 = 0.20$$

3c) $\mathbb{P}(Y_2 = y)$



Using the table for Y_2 :

$$F_{Y_2}(6) = \mathbb{P}(Y_2 \leq 6) = 0 + 0 + 0.01 + 0.04 + 0.11 + 0.20 + 0.25$$

$$= 0.61$$

Because $Y_2 \sim \text{Bin}(n=10, p=0.6)$, we expect 60% of questions to be answered correctly on average in Class 2. So it makes sense that a score of 6/10 is most probable, assuming all children answer all questions independently with $\mathbb{P}(\text{correct}) = 0.6$.

3d) Let $A = \{\text{script has } 7/10\}$

$B_x = \{\text{script is from class } x\}$ for $x = 1, 2, 3$

– defined on the sample space $\Omega = \{\text{all scripts}\}$.

Given information: $\mathbb{P}(B_1) = \mathbb{P}(B_2) = \mathbb{P}(B_3) = \frac{1}{3}$.

Using the table for Y_1, Y_2, and Y_3, we have:

$\mathbb{P}(A \mid B_1) = \mathbb{P}(Y_1 = 7) = 0.01$

$\mathbb{P}(A \mid B_2) = \mathbb{P}(Y_2 = 7) = 0.22$

$\mathbb{P}(A \mid B_3) = \mathbb{P}(Y_3 = 7) = 0.06$

We want to find $\mathbb{P}(A) = \mathbb{P}(\text{randomly selected script has } 7/10)$.

Use the Partition Theorem:

$$\mathbb{P}(A) = \mathbb{P}(A \mid B_1)\mathbb{P}(B_1) + \mathbb{P}(A \mid B_2)\mathbb{P}(B_2) + \mathbb{P}(A \mid B_3)\mathbb{P}(B_3)$$
$$= 0.01 * \tfrac{1}{3} + 0.22 * \tfrac{1}{3} + 0.06 * \tfrac{1}{3}$$

$\Rightarrow \underline{\underline{\mathbb{P}(A) = 0.097}}$.

3e) We seek $\mathbb{P}(B_2 \mid A) = \dfrac{\mathbb{P}(A \mid B_2)\,\mathbb{P}(B_2)}{\mathbb{P}(A)}$

$$= \frac{0.22 * 1/3}{0.097} \quad \text{using (b)}$$

$\Rightarrow \underline{\underline{\mathbb{P}(B_2 \mid A) = 0.759}}$

– probability a last script with score 7/10 comes from Class 2.

BONUS QUESTION

5) Let $\Omega = \{\text{all possible pairs of (class 2 score, class 3 score)}\}$.

Define $H = \{Y_2 > Y_3\} = \{\text{pair has higher class 2 score than class 3 score.}\}$

$S_y = \{Y_3 = y\} = \{\text{class 3 score is } y\}$ for $y = 0, 1, \ldots, 10$.

Then $S_0, S_1, \ldots, S_{10}$ form a partition of Ω, because every pair must satisfy one and only one of these events.

Using the Partition Theorem:

$$\mathbb{P}(H) = \sum_{y=0}^{10} \mathbb{P}(H \mid S_y)\,\mathbb{P}(S_y)$$

Equivalently: $\mathbb{P}(Y_2 > Y_3) = \sum_{y=0}^{10} \mathbb{P}(Y_2 > Y_3 \mid Y_3 = y)\,\mathbb{P}(Y_3 = y)$

$$\Rightarrow \underline{\underline{\mathbb{P}(Y_2 > Y_3) = \sum_{y=0}^{10} \mathbb{P}(Y_2 > y)\,\mathbb{P}(Y_3 = y)}} \quad \text{as stated.}$$

Note that $\mathbb{P}(Y_2 > Y_3 \mid Y_3 = y) = \mathbb{P}(Y_2 > y)$, because once we know that $Y_3 = y$, there is no further influence of Y_3 on whether $Y_2 > Y_3$ or not.

Using the table and running through the y-values with $\mathbb{P}(Y_3 = y) > 0$ according to the table:

$$\mathbb{P}(Y_2 > Y_3) = 0.01 * (0.22 + 0.12 + 0.04 + 0.01) \quad (y = 6)$$
$$+ 0.06 * (0.12 + 0.04 + 0.01) \quad (y = 7)$$
$$+ 0.19 * (0.04 + 0.01) \quad (y = 8)$$
$$+ 0.39 * 0.01 \quad (y = 9)$$

$\Rightarrow \underline{\underline{\mathbb{P}(Y_2 > Y_3) = 0.0275}}$ No, it is not likely that a child in class 2 will do better than a child in class 3.